Earth and Planetary Science Letters, 105 (1991) 492–499
Elsevier Science Publishers B.V., Amsterdam

[DT]

Baked contact test demonstrates primary nature of dominant (N1) magnetisation of Nipissing intrusions in Southern Province, Canadian Shield

Kenneth L. Buchan

Geological Survey of Canada, 601 Booth St., Ottawa, Ont. K1A 0E8, Canada

Received December 20, 1990; revised version accepted May 29, 1991

ABSTRACT

Nipissing diabase sills, cone sheets and dykes define a major episode of igneous intrusion in the Southern Province of the Canadian Shield at about 2200 Ma. The dominant (N1) remanence of these intrusions has been established in several palaeomagnetic studies and, hence, is a key tie point for the North American apparent polar wander path. Yet the age of this magnetisation remains very controversial. Positive baked contact tests reported for the N1 remanence have often been dismissed as inconclusive and the N1 palaeopole attributed to overprinting during the Hudsonian Orogeny (1900–1800 Ma) due to its position on a widely accepted polar wander path for North America.

Previous baked contact studies utilise sediments of the Huronian Supergroup as host rock. The remanence of these sediments is often difficult to interpret. In this study a positive baked contact test is described for an N1 Nipissing sill or cone sheet using stably magnetized mafic dykes of the Matachewan swarm as host. Thus, the N1 remanence is primary and no longer supports the time calibration of polar wander paths which assume that it is a 1900–1800 Ma overprint.

1. Introduction

The age of the northerly upward N1 magnetisation which dominates 2200 Ma Nipissing diabase intrusions (sills, cone sheets and dykes) of the Southern Province of the Canadian Shield [1–4] is critical to any interpretation of the early Proterozoic polar wander path for North America [5]. Nevertheless, the primary or secondary nature of this remanence has long been controversial (e.g. [5–9]). In the present study, a baked contact test was conducted between an N1 Nipissing sill or cone sheet and older Matachewan diabase dykes in order to resolve this controversy.

On a widely used version of the polar wander path for North America (e.g. [8,10–12]) the N1 palaeopole falls at 1900–1800 Ma, suggesting that the N1 remanence is secondary and was acquired during the Hudsonian Orogeny, about 300 or 400 Ma after Nipissing intrusion. According to this interpretation, the westerly down N2 remanence observed at other Nipissing localities, is primary. The N2 palaeopole lies near 2100–2000 Ma on this polar wander path. Indeed, a positive baked contact test has been reported for N2 sills of the Cobalt area [10].

On the other hand, Symons [13], Morris [14] and Stupavsky and Symons [4] have reported positive baked contact tests for the N1 Nipissing remanence, implying that this magnetisation is primary and, hence, about 2200 Ma old. These tests, carried out with host sediments of the Early Aphebian Huronian Supergroup, have not been accepted as reliable by many workers (e.g. [5,6,8,10,11,12,15,16]). The scepticism is due to the nature of the palaeomagnetic data described in each N1 baked contact study.

2. Previous baked contact tests

Problems with two of the N1 baked contact tests for Nipissing sills [13,14] have been discussed in detail by McGlynn and Irving [6].

Symons [13] reported a northerly upward N1 magnetisation ($D = 28.6°$, $I = -44.6°$, $\alpha_{95} = 7.2°$) for a Nipissing sill in the Cobalt area, and a

0012-821X/91/$03.50 © 1991 – Elsevier Science Publishers B.V. All rights reserved

significantly different direction ($D = 35.3°$, $I = -74.1°$, $\alpha_{95} = 14.6°$) for unbaked Cobalt Group sediments of the Huronian Supergroup some distance away. Baked sediments of the upper slate member of the Cobalt Group adjacent to the sill carry very scattered remanence directions ($D = 41.4°$, $I = -30.4°$, $\alpha_{95} = 48.3°$) whose large circle of 95% confidence overlaps the circles of confidence for both the Nipissing sill and the unbaked Cobalt Group sediments. Symons concluded that "the remanence of the upper slate member is much more akin to that of the Nipissing sill than to that of the Cobalt sediments". Nevertheless, this result has not convinced many palaeomagnetists.

Morris [14] reported a positive baked contact test for a reversed (i.e., down to the southwest) component in a Nipissing intrusion near Elliot Lake. The magnetisation of the two Nipissing sampling sites is described as "extremely complex". One site is in a coarse-grained phase. Below 550°C successive thermal demagnetisation steps remove oppositely directed components with overlapping blocking temperature spectra. The mean direction is $D = 222°$, $I = 19°$, $\alpha_{95} = 7°$. Above 550°C the magnetisation at this site is usually unstable, although a few specimens have a steeper direction ($D = 231°$, $I = 49°$, $\alpha_{95} = 10°$). A second site is in fine-grained diabase. Once again, sequential demagnetisation steps remove remanence components of opposing polarity ($D = 223°$, $I = 52°$, $\alpha_{95} = 7°$). Gowganda argillite of the Huronian Supergroup close to the intrusion carries a southwesterly downward direction of magnetisation ($D = 225°$, $I = 44°$, $\alpha_{95} = 5°$), while unbaked argillite farther from the intrusion carries a significantly different direction ($D = 338°$, $I = 68°$).

It is also uncertain whether the southwesterly down remanence of Morris [14] represents the N1 component of Nipissing sills [7] . In a later publication, Morris [17] plotted all Nipissing palaeopoles for the Cobalt Plate and Blind River region of the Southern Province, including the Nipissing virtual geomagnetic pole for the baked contact test of Morris [14] near Elliot Lake in the Blind River region. He demonstrated that the N1 palaeopoles form two groupings, one for the Blind River area and the other for the Cobalt Plate, and proposed that the difference is due to relative rotation of the two regions after remanence acquisition. The virtual pole for the Elliot Lake baked contact test of Morris [13], however, falls far from the grouping of N1 poles for the Blind River region. Indeed, it is almost as close (~ 55°) to the N2 Blind River pole as it is (~ 45°) to the N1 Blind River grouping. Therefore, it is questionable whether the Elliot Lake baked contact actually tests an N1 remanence.

Finally, Stupavsky and Symons [4] observed a northerly upward N1 component in two baked Huronian sites of the Gowganda Formation southwest of Lake Temagami. One site straddles the Nipissing–Gowganda contact and includes two Nipissing diabase cores. Unfortunately, the sampling profile at this locality does not extend into unbaked Gowganda sediments and other nearby sites in the Gowganda sediments are unstably magnetised. Thus, the baked contact test is inconclusive even though, at a distance of about four kilometres or more from the Nipissing sill site, a steeply downward, pre-Nipissing remanence is obtained [4].

3. Geology and palaeomagnetic sampling

In the present study a baked contact test is described for the N1 magnetisation of a Nipissing sill or cone sheet at Everett Lake, four kilometres northeast of the town of Gowganda in the Cobalt Plate (Fig. 1). The locality was selected for several reasons.

(1) The palaeomagnetism of Nipissing intrusions in this area is well documented [18] although no baked contact test has been reported.

(2) The Nipissing sill at the sampling locality carries an unambiguous northerly upward N1 remanence which cannot be confused with the other remanence directions that have been reported for Nipissing intrusions.

(3) Mafic dykes of the Matachewan swarm provide an ideal host rock for the baked contact study. They are readily identified by their northerly trend and by a stable magnetisation which is directed to the south-southwest (e.g. [19–21]). This magnetisation is known to be primary on the basis of a baked contact test [21] and is easily distinguished from the N1 direction of Nipissing sills.

In the study area of Fig. 1, Matachewan diabase dykes intrude Archean volcanic and plutonic rocks. The Matachewan dyke swarm has been precisely dated at $2452 + 3/-2$ Ma using the U-Pb zircon technique [22]. Sediments of the Gowganda Formation of the Huronian Supergroup rest unconformably on the Archean basement and are little deformed. They outcrop immediately south of the sampling area shown in Fig. 1. Nipissing sills and cone sheets intrude both Archean rocks and Huronian sediments. A nearby Nipissing sill has been dated at 2217 ± 1.6 Ma by the U-Pb baddeleyite technique [23].

Nipissing intrusions of the Gowganda area carry either a northerly upward N1 remanence or a steeply downward magnetisation [18]. The latter has been interpreted as either a present field remanence [18] or as an N2 component [10]. To establish the direction of magnetisation of the sill or cone sheet near Everett Lake, samples were collected at sites A and B of Fig. 1. To investigate the effect of sill intrusion on the magnetisation of Matachewan dykes, sampling sites C, D, E and F were collected in north-trending dykes at horizontal distances of about 90, 300, 800 and 1000 m, respectively, from the sill contact. The sill–dyke contact itself, however, is not exposed. Nor is there sufficient outcrop to allow collection of a continuous profile of dyke samples with increasing distance from the sill.

Identification of dykes at sites C, D, E and F as part of the Matachewan swarm is based on dyke trend and, in the case of sites D, E and F located relatively far from the Nipissing intrusion, on palaeomagnetic direction (see below) . Three dyke sites (C, D and E) that were examined petrographically consist of partly altered pyroxene quartz diabase similar to Matachewan dykes elsewhere in the region (K. Card, pers. commun.). However, petrographic characteristics alone do not yield a unique correlate since Matachewan dykes are not readily distinguished from Nipissing sills and dykes or from northeast-trending Preissac dykes on the basis of petrography [24].

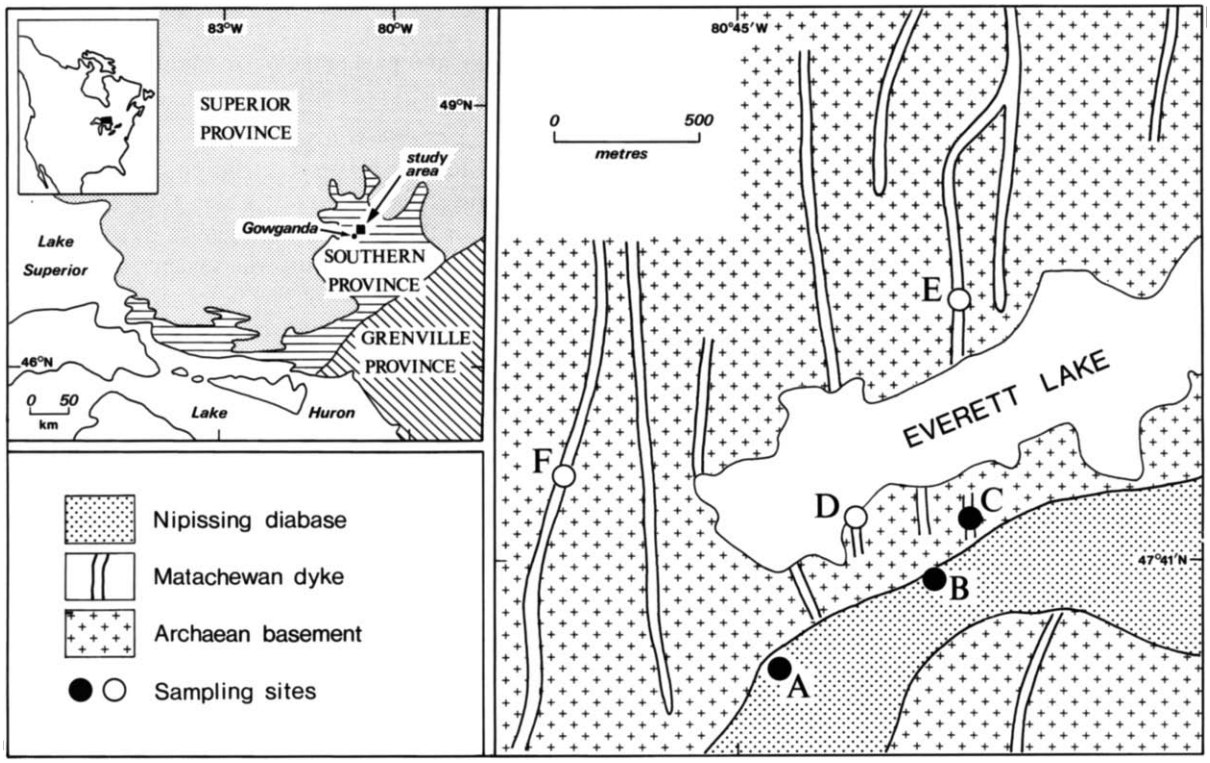


Fig. 1. Geology and sampling localities near Everett Lake in the Southern Province of the Canadian Shield. Sampling sites that carry an N1 magnetisation are shown as closed symbols, those that have a Matachewan direction as open symbols. Geology is modified after Ontario Division of Mines Map 2349 (Haultain and Nicol Townships), 1976.

Five samples (either drilled cores or blocks) were oriented by solar and/or magnetic compass at each site and two or more cylindrical specimens (2.5 cm diameter, 2.2 cm length) were cut from each. Two specimens per sample were demagnetised in peak alternating fields (AF's) of 2.5, 5.0, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90 and 100 mT and measured with a Schoensted spinner magnetometer. One specimen per site was thermally demagnetised in peak temperatures of 200, 300, 400, 450, 500, 530, 550, 560, 570, 580 and 590°C using a Schoensted furnace modified to control temperatures to $\pm 5°$.

Magnetisation directions were averaged from two specimens per sample at each AF-cleaning step. A mean direction was then calculated for each sampling site using the field corresponding to the best grouping of directions after a stable end point was attained.

As Huronian sediments in the sampling area are relatively flat lying, no structural correction has been applied to the palaeomagnetic directions. Similarly, the discussion of other Nipissing results is limited to those from areas where Huronian rocks are little deformed and metamorphosed.

4. Results

All sampling sites yield stable directions of magnetisation after AF or thermal cleaning (Fig. 2) . Analysis of orthogonal component plots for each specimen (e.g. Fig. 3) indicate that unstable or viscous components eliminated in the early stages of demagnetisation have directions close to that of the present earth's field or are highly scattered.

At Nipissing sites A and B (Fig. 1) a typical northerly upward N1 magnetisation direction is attained upon AF demagnetisation in fields of 15–50 mT (Figs. 2a, 3a) . The N1 end point is stable and similar to that observed for other Nipissing sills of the Gowganda area [18]. Unblocking temperatures of the stable component fall in a narrow range below 580°C, the Curie temperature of magnetite, indicating that this component is carried by low-titanium titanomagnetite. Site mean directions are given in Table 1 and plotted in Fig. 4. For comparison the mean N1 direction reported previously from the Gowganda area [18] is also shown in Fig. 4.

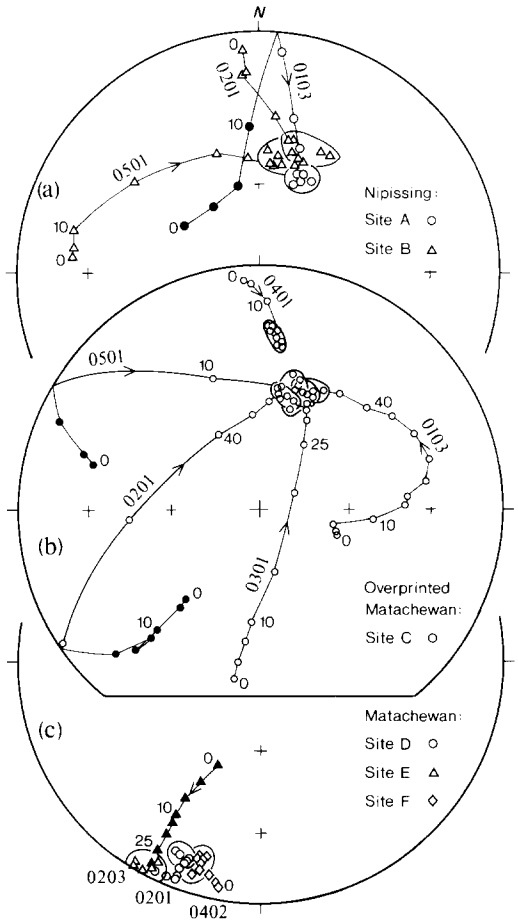


Fig. 2. Palaeomagnetic results obtained on AF demagnetisation for specimens from (a) Nipissing sill sites *A* and *B*, (b) baked Matachewan dyke site *C* close to the sill and (c) unbaked Matachewan dyke sites *D*, *E* and *F* farther from the sill contact. Directions are plotted on equal area nets with closed (open) symbols indicating down (up) directions.

At a horizontal distance of about 90 m from the Nipissing intrusion, Matachewan dyke site C (Fig. 1) carries a stable N1 magnetisation after demagnetisation in AF's of 15–60 mT (Figs. 2b, 3b) . Once again the N1 remanence is very stable, has unblocking temperatures indicative of low-titanium titanomagnetite and is indistinguishable in direction from that observed in nearby Nipissing sills [18].

Farther from the Nipissing intrusion (at horizontal distances of about 300, 800 and 1000 m), Matachewan dyke sites D, E and F (Fig. 1) carry shallow south-southwesterly directions of magnetisations that are attained in AF's of 15–30 mT

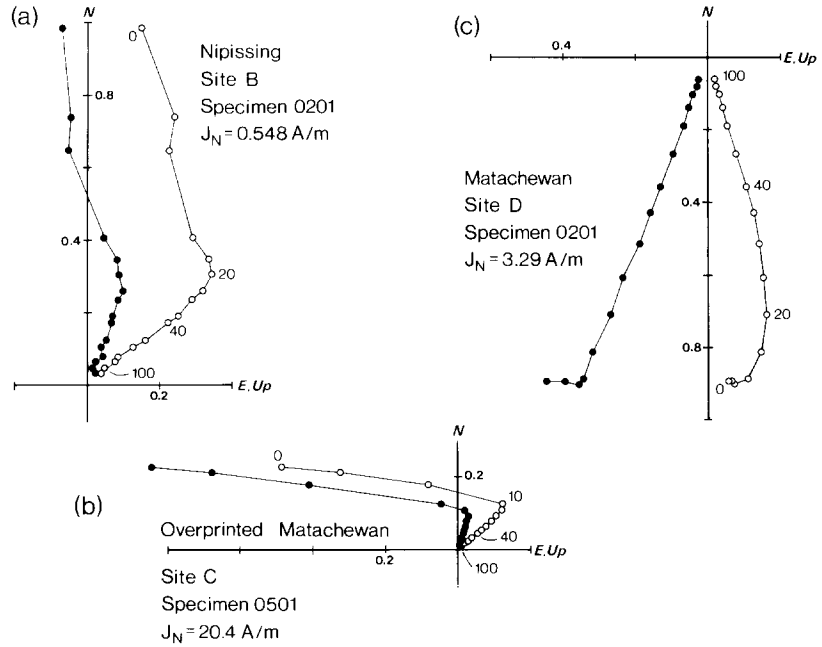


Fig. 3. Orthogonal component plots for specimens from (a) Nipissing sill site *B*, (b) Matachewan dyke site *C* carrying an overprint due to the nearby Nipissing sill and (c) Matachewan dyke site *D* carrying its original remanence. Closed symbols indicate projection onto the horizontal plane; open symbols indicate projection onto the up and north vertical plane.

(Figs. 2c, 3c) and unblocking temperatures close to the Curie temperature of magnetite (580°C) . As noted above, this direction is typical of Matachewan dykes throughout much of the southern portion of the Superior Province and has been shown to be primary.

As illustrated by the data of Fig. 4, the Matachewan dyke at site C, close to the Nipissing intrusion, was overprinted by an N1 remanence at the time of Nipissing intrusion. More distant Matachewan dyke sites D, E and F, however, escaped overprinting. This result constitutes a positive baked contact test for the N1 magnetisation of the Nipissing diabase.

A primary age for the N1 remanence of the Nipissing intrusion at this locality is supported by the consistency in the blocking temperatures of the stable components at all sampling sites. As blocking temperatures at each site fall in a narrow range near 580°C indicating that low-titanium titanomagnetite carries the stable remanence, it is difficult to envisage a mechanism that would com-

TABLE 1

Site mean palaeomagnetic directions for Gowganda baked contact test

Site	*N*	Unit	Dir	AF	D	I	k	α_{95}
A	5	Nipissing sill	N1	80	24.4	−52.0	354	4.1
B	5	Nipissing sill	N1	80	16.1	−46.2	86	8.3
C	5	baked Matachewan dyke	N1	80	10.2	−43.2	65	9.6
D	5	unbaked Matachewan dyke	M	30	206.9	−11.3	217	5.2
E	5	unbaked Matachewan dyke	M	80	203.0	−9.1	86	8.3
F	5	unbaked Matachewan dyke	M	50	206.3	−17.4	102	7.6

N = number of samples per site; Dir = direction of stable magnetisation as described in text; AF = alternating field; D = declination in degrees; I = inclination in degrees; k = precision parameter; α_{95} = radius of circle of 95% confidence in degrees.

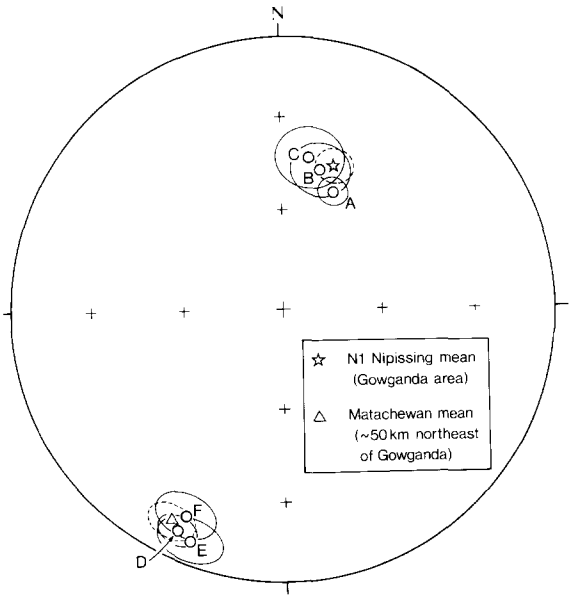


Fig. 4. Mean directions of magnetisation and circles of 95% confidence for each sampling site. The star with its dashed circle of confidence is the mean N1 remanence direction reported by Symons and Londry [18] for the six Nipissing sites of the Gowganda area. The triangle with its dashed circle of confidence is the mean direction reported by Buchan et al. [21] for sixteen Matachewan dykes about 50 km northeast of the study area.

pletely overprint low-titanium titanomagnetite of N1 sites but not of nearby sites with a Matachewan remanence direction.

5. Discussion

The positive baked contact test described herein demonstrates that the N1 remanence of the Nipissing sills in the Gowganda area is primary. It was acquired at the time of Nipissing intrusion about 2200 Ma ago, rather than as a secondary overprint during the Hudsonian Orogeny 300–400 Ma later. Thus, N1 palaeopoles which fall in the Pacific near the west coast of South America should no longer be used to support widely accepted polar wander paths for North America that pass through this area at 1900–1800 Ma (e.g. [8,10–12] and Fig. 5).

Most palaeomagnetists have constructed Precambrian polar wander paths for North America by incorporating palaeopoles from across the Canadian Shield. Roy [7], however, has demonstrated that Precambrian poles are usually too poorly dated to establish whether the different structural provinces have occupied their present relative positions since that time. Hence, it may be more appropriate to consider polar wander for individual elements of the Shield—in this case, for the Southern Province northeast of Lake Huron and the adjacent portion of the Superior Province (Fig. 1) which are known to have remained largely intact since intrusion of the Matachewan dyke swarm 2450 Ma ago.

Although the Southern–Superior region has one of the most extensive palaeomagnetic data bases of any in the Canadian Shield, there are few well-dated poles for the 1900–1800 Ma period. None are located in the Pacific Ocean near the west coast of South America, with the exception of the N1 poles of the Nipissing diabase (Fig. 5) and poles reported from the 2150 Ma Preissac dyke swarm. However, the N1 Nipissing poles are shown in this study to date from the time of intrusion at about 2200 Ma and the age of the Preissac magnetisation is unknown, eliminating

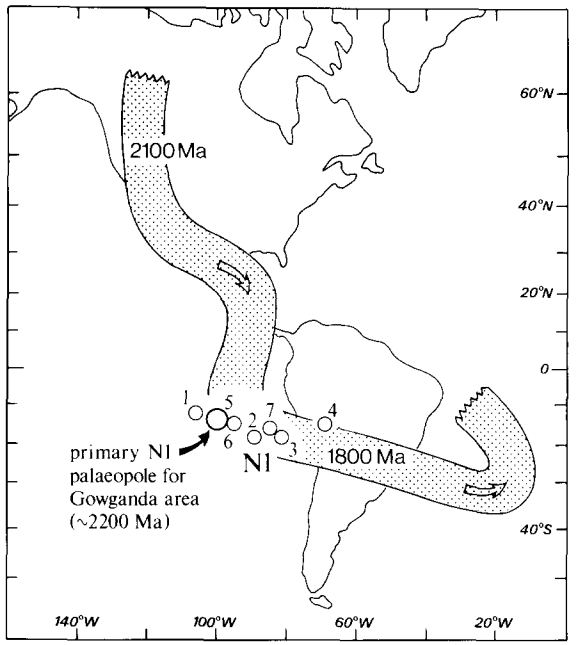


Fig. 5. N1 Nipissing diabase palaeopoles (circles: pole 1, [13]; pole 2, [1]; pole 3, [2] recalculated in [17]; pole 4, [3]; pole 5, [18]; pole 6, [10]; pole 7, [4]) are compared with a recent example of the apparent polar wander path for North America (shaded curve, after [8]). Nipissing pole 5 (large circle) of the Gowganda area is established as primary in this study on the basis of a baked contact test.

evidence for a Southern–Superior polar wander path passing through this area at 1900–1800 Ma. Indeed, there is little reliable data from the Southern–Superior region to support the 2100–1700 Ma portion of the polar wander path which is often attributed to the Canadian Shield as a whole (e.g. [8,10–12] and Fig. 5).

For example, the age of the N2 Nipissing palaeopoles which fall near 2200–2100 Ma on the standard polar path is not well established. Therefore, the relative age of N1 and N2 palaeopoles is uncertain and it remains unclear whether the path should be drawn from north to south between N2 and N1 poles or from south to north between N1 and N2. Indeed, a third remanence (called N3) which occurs in Nipissing sills in one locality has recently been recognized as primary [25] and its age relationship to N1 and N2 is still under study. It has been proposed that distinct directions of magnetisation in Nipissing intrusions may indicate different ages of emplacement (e.g. [4,25]).

6. Conclusions

This study demonstrates that the N1 remanence from Nipissing diabase of the Southern Province is primary. Therefore, when constructing apparent polar wander paths for this region, or for North America as a whole, the N1 palaeopole should no longer be assumed to represent overprinting at 1900–1800 Ma, as has often been the case. Indeed, until more reliably dated palaeopoles for this period have been obtained from the Southern or adjacent Superior Province, it should not be assumed that the widely accepted North American polar wander path of Fig. 5 is applicable to the Southern–Superior region.

Acknowledgements

I wish to thank M. Pyne for assistance in the field and laboratory, K. Card for discussions of the geology of the Nipissing diabase and Matachewan dykes, and reviewers who have suggested improvements to the paper. Geological Survey of Canada contribution no. 36690.

References

1 D.T.A. Symons, Paleomagnetism of the Nipissing diabase, Cobalt area, Ontario, Can. J. Earth Sci. 7, 86–90, 1970.
2 D.T.A. Symons, A paleomagnetic study of the Nipissing Diabase, Blind River-Elliot Lake area, Ontario, Geol. Surv. Can. Pap. 70-63, 19–30, 1971.
3 J.P. Patel and H.C. Palmer, Magnetic and paleomagnetic studies of the Nipissing Diabase, Lake Matinenda area, Ontario, Can. J. Earth Sci. 11, 353–361, 1974.
4 M. Stupavsky and D.T.A. Symons, Extent of Grenvillian remanence components in rocks of the Southern Province, Can. J. Earth Sci. 19, 698–708, 1982.
5 E. Irving and J.C. McGlynn, On the coherence, rotation and paleolatitude of Laurentia in the Proterozoic, in: Precambrian Plate Tectonics, A. Kroner, ed., pp. 561–598, Elsevier, London, 1981.
6 J.C. McGlynn and E. Irving, Horizontal motions and rotations in the Canadian Shield during the Early Proterozoic, in: Proterozoic Basins of Canada, F.H.A. Campbell, ed., Geol. Surv. Can. Pap. 81-10, 183–190, 1981.
7 J.L. Roy, Paleomagnetism of the North American Precambrian: a look at the data base, Precambrian Res. 19, 319–348, 1983.
8 H. Hyodo and D.J. Dunlop, Multicomponent magnetizations in the Nipissing diabase near Temagami, Ontario, and structural and thermal history of the Grenville Front, Can J. Earth Sci. 26, 467–478, 1989.
9 J.D.A. Piper, Palaeomagnetism and the Continental Crust, 434 pp., Open University, Milton Keynes, 1987.
10 J.L. Roy and P.L. Lapointe, The paleomagnetism of Huronian red beds and Nipissing diabase; Post-Huronian igneous events and apparent polar path for the interval −2300 to −1500 Ma for Laurentia, Can. J. Earth Sci. 13, 749–773, 1976.
11 E. Irving, Paleopoles and paleolatitudes of North America and speculations about displaced terrains, Can. J. Earth Sci. 16, 669–694, 1979.
12 D.J. Dunlop and L.D. Schutts, Proterozoic magnetic overprinting and tectonics of the Laurentian Shield, Geol. Soc. Am. Mem. 161, 35–45, 1983.
13 D.T.A. Symons, Paleomagnetism of Precambrian rocks near Cobalt, Ontario, Can. J. Earth Sci. 4, 1161–1169, 1967.
14 W.A. Morris, A positive contact test between Nipissing diabase and Gowganda argillites, Can. J. Earth Sci. 16, 607–611, 1979.
15 E. Irving, A. Davidson and J.C. McGlynn, 1984, Paleomagnetism of gabbros of the Early Proterozoic Blachford Lake Intrusive Suite and the Easter Island Dyke, Great Slave Lake, NWT: possible evidence for the earliest continental drift, Geophys. Surv. 7, 1–25, 1984.
16 L.D. Schutts and D.J. Dunlop, Proterozoic magnetic overprinting of Archean rocks in the Canadian Superior Province, Nature 291, 642–645, 1981.
17 W.A. Morris, Fault block rotations in the Southern Province as defined by paleomagnetism of the Nipissing diabase, Can. J. Earth Sci. 18, 1755–1757, 1981.
18 D.T.A. Symons and J.W. Londry, Tectonic results from paleomagnetism of the Aphebian Nipissing Diabase at Gowganda, Ontario, Can. J. Earth Sci. 12, 940–948, 1975.
19 E. Irving and A.J. Naldrett, Paleomagnetism in Abitibi Greenstone Belt, and Abitibi and Matachewan diabase dikes: Evidence of the Archean geomagnetic field, J. Geol. 85, 157–176, 1977.

20 M.P. Bates and H.C. Halls, Magnetic polarity domains in the Matachewan dyke swarm associated with the Kapuskasing Structural Zone, Can. J. Earth Sci., in press, 1990.
21 K.L. Buchan, D.J. Neilson and C.J. Hale, Relative age of Otto stock and Matachewan dykes from paleomagnetism and implications for the Precambrian polar wander path, Can. J. Earth Sci., in press, 1989.
22 L.M. Heaman and J. Tarney, U-Pb baddeleyite ages for the Scourie dyke swarm, Scotland: evidence for two distinct intrusion events, Nature 340, 705–708, 1989.
23 F. Corfu and A.J. Andrews, A U-Pb age for mineralized Nipissing diabase, Gowganda, Ontario, Can. J. Earth Sci. 23, 107–109, 1986.
24 K.D. Card, Progress report on regional geological synthesis, central Superior Province, Part 1, Reconnaisance investigations and studies of igneous intrusions, Geol. Surv. Can. Pap. 81-1A, 78–82, 1981.
25 K.L. Buchan, K.D. Card and F.W. Chandler, Multiple ages of Nipissing diabase intrusion: paleomagnetic evidence from the Englehart area, Ontario, Can. J. Earth Sci. 26, 427–445, 1989.